

Monopolistic Competition & Oligopoly

Chapter 24

Theory of Monopolistic Competition

- There are many sellers and buyers
- Each firm in the industry produces and sells a slightly differentiated product. Product differentiation may be real or imagined.
- There is easy entry and exit. There are no barriers to entry or exit.

The Nature of Monopolistic Competition

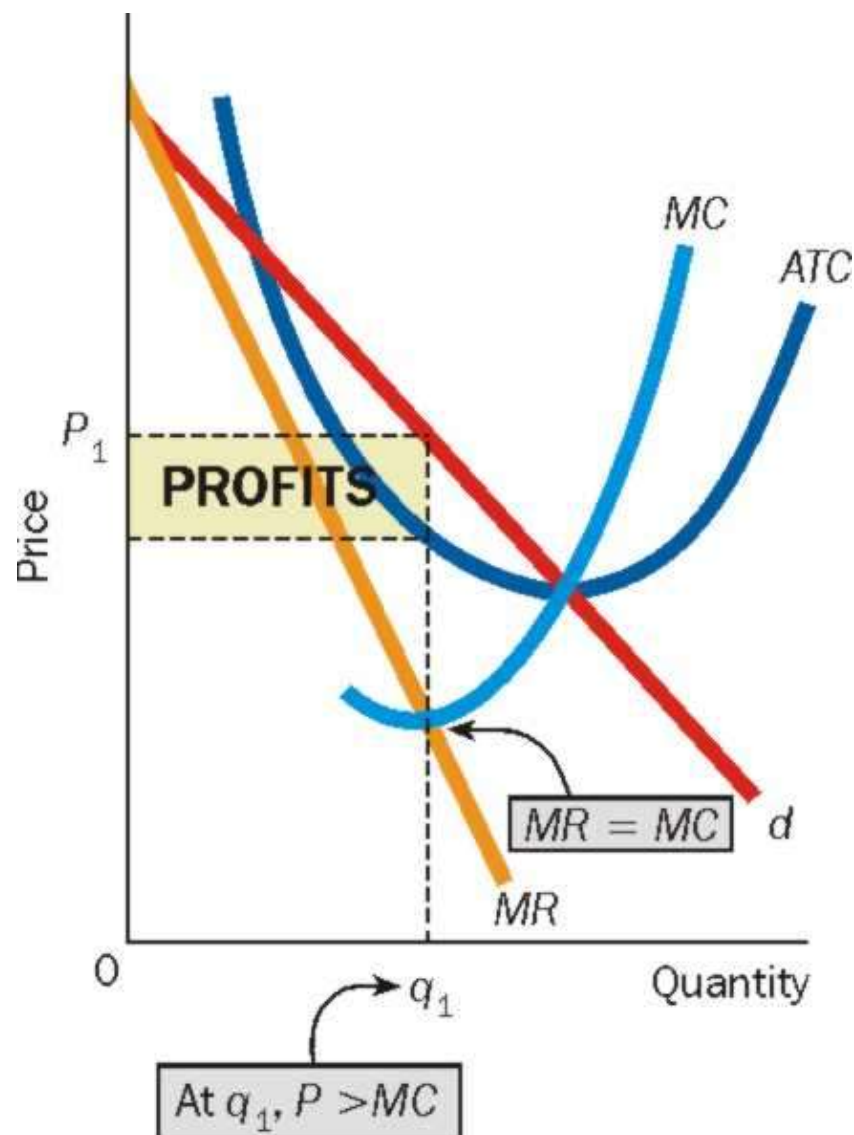
- Demand curve: Like perfect competitive firms, it has many rivals, but unlike the competitive firm its rivals do not sell exactly the same product. So the demand curve is not perfectly elastic.

Again, Monopoly produces a good for which there are no substitutes. But the monopolistic competitive firm sells a good which has close substitutes. So the demand curve is much more elastic than the monopoly firm's demand curve.

- In perfect competition $P=MC$.
- In the monopolistic competitor $P>MR$.

The Monopolistic Competitive Output and Price

The monopolistic competitor produces that quantity of output for which $MR=MC$. This is Q_1 in the exhibit. It charges the highest price consistent with the quantity, which is P_1 .

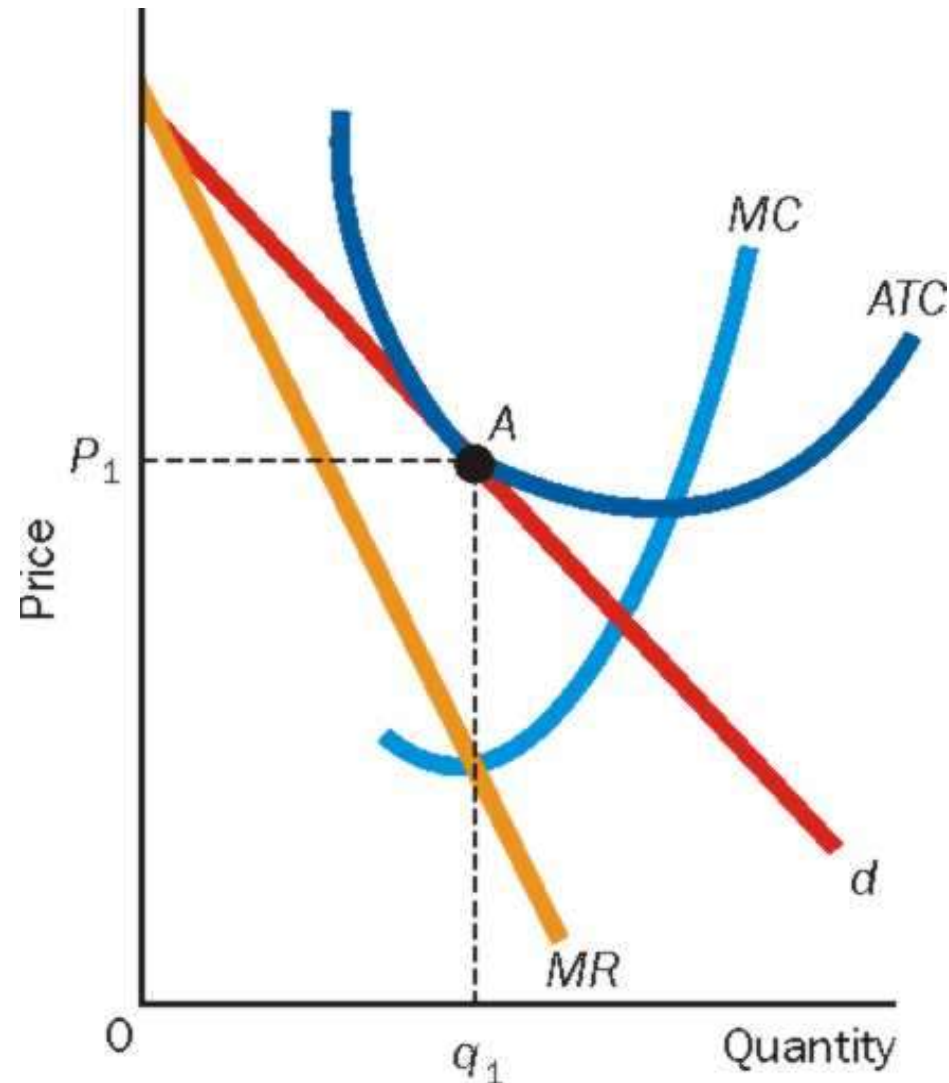


Will There be Profits in the Long Run?

- If firms in the industry are earning profits, new firms will enter the industry and reduce the demand that each firm faces.
- Eventually, competition will reduce economic profits to zero in the long run.

Monopolistic Competition in the Long Run

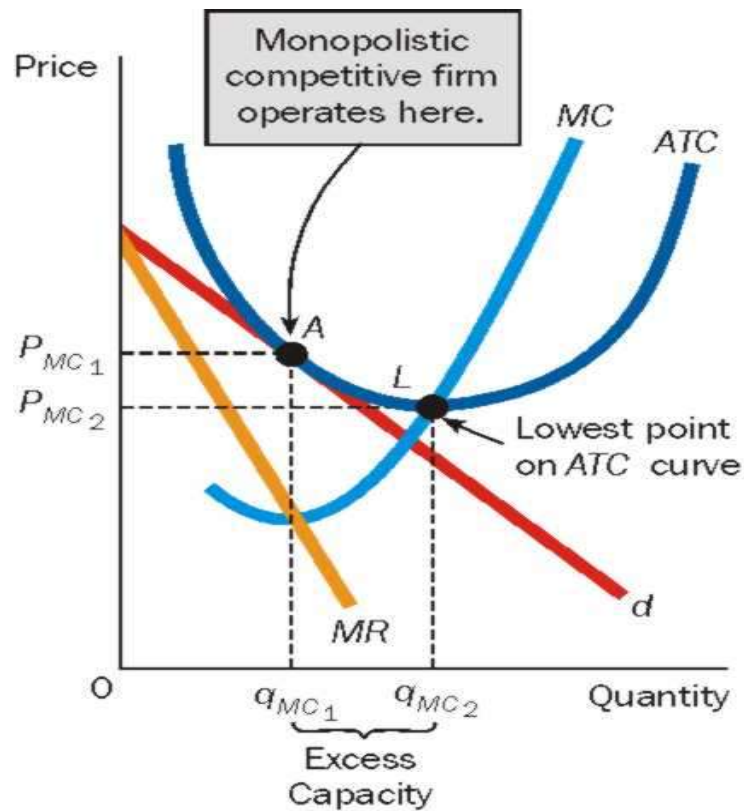
Because of easy entry into the industry, there are likely to be zero economic profits in the long run for a monopolistic competitor. In other words, $P=ATC$



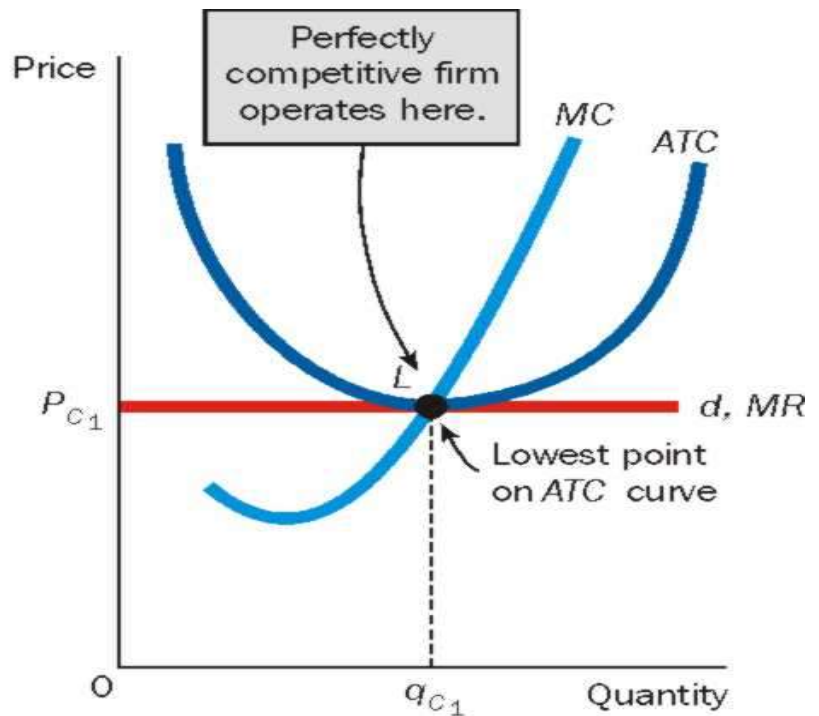
Excess Capacity: What is it, and Is it Good or Bad?

- Excess capacity theorem: in equilibrium a monopolistic competitor will produce an output smaller than the one that would minimize its unit cost of production.
- In long-run equilibrium, when the monopolistic competitor earns zero economic profits, it is not producing the quantity of output at which average total costs are minimized for the given scale of plant.
- In long-run equilibrium, the perfectly competitive firm produces the quantity of output at which unit costs are minimized.

A Comparison of Perfect Competition and Monopolistic Competition



(a)
Monopolistic Competition



(b)
Perfect Competition

Excess Capacity or underutilization of plant size.

- The perfectly competitive firm produces a quantity of output consistent with lowest unit costs. The monopolistic competitor does not. If it did, it would either produce q_{MC_2} , instead of q_{MC_1} . The monopolistic competitor is said to underutilize its plant size or to have excess storage capacity.

Demand Curves and Advertising

- In short, the monopolistic competitor operates at excess capacity as a consequence of its downward sloping demand curve.
- Its downward sloping demand curve is a consequence of differentiated products.
- Firms sometimes use advertising to try to differentiate their products from their competitor's products.
- As $P > MC$, resource allocative efficiency is not achieved.
- As monopolistic competitor operates at excess capacity (not at the lowest point of ATC) productive efficiency is also not achieved.

Oligopoly: Assumptions and Real-World Behavior

- There are few sellers and many buyers. Firms are interdependent: each firm is aware that its actions influence the other firms and that the actions of the other firms affect it.
- Firms produce and sell either homogeneous or differentiated products.
- There are significant barriers to entry. Economies of scale, patent rights, exclusive control of an essential resource and legal barriers act as barriers of entry.
- Concentration Ratio: The percentage of industry sales accounted for by a set number of firms in the industry.

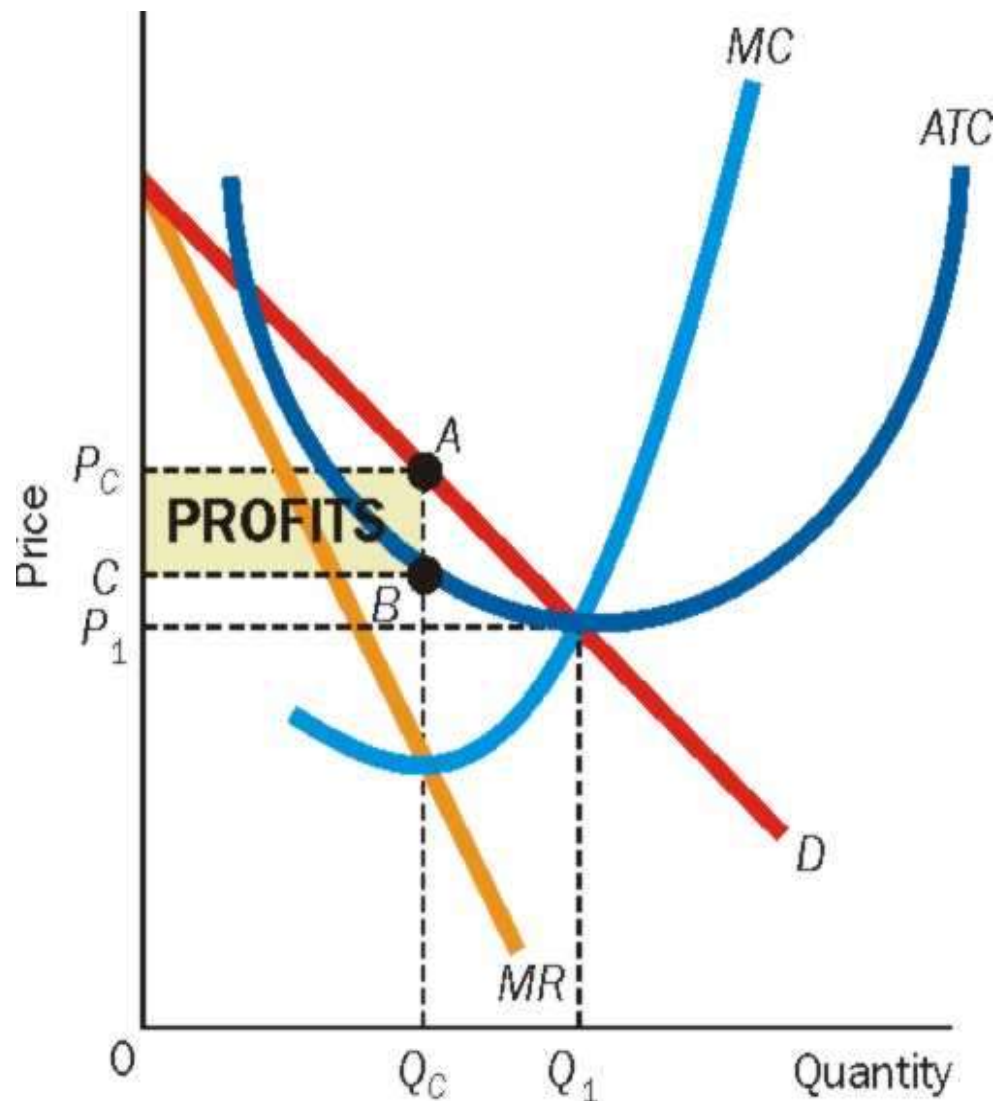
Price and Output under Oligopoly

- Cartel Theory: Oligopolists in an industry act as if there were only one firm in the industry.
- A Cartel is an organization of firms that reduces output and increases price in an effort to increase joint profits.

Problem of forming cartel: Getting the sellers of an industry together to form a cartel can be costly. Each potential member has an incentive to be a free rider, to stand by and take a free ride from the actions of others.

The Benefits of Being Members of a Cartel

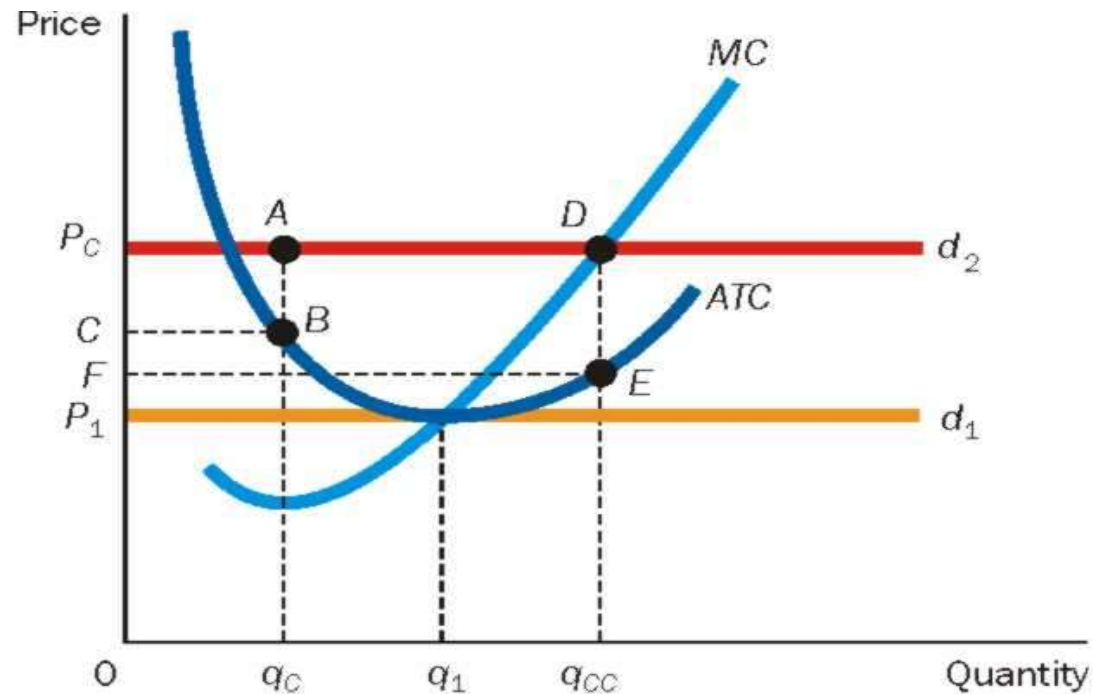
We assume the industry is in long-run equilibrium, producing Q_1 , and charging P_1 . There are no profits. A reduction in output to Q_C through the formation of a cartel raises price to P_C and brings profits of CP_CAB



Problems with Cartels

- High profits will provide an incentive for firms from outside the industry to join the industry.
- After the cartel agreement is made, cartel members have an incentive to cheat on the agreement.
- If a firm cheats on the cartel agreements and other firms do not, then the cheating firm can increase its profits. Of course if all firms cheat, the cartel members are back where they started at: no cartel agreements and at the original price.

Benefits of Cheating in a Cartel Agreement



Quantity firm produces when it adheres to cartel agreement;
profits = CP_CAB

Quantity firm produces when there is no cartel agreement;
profits = 0

Quantity firm produces when it cheats on the cartel agreement, but other firms do not;
profits = FP_CDE

The problem of formulating cartel policy

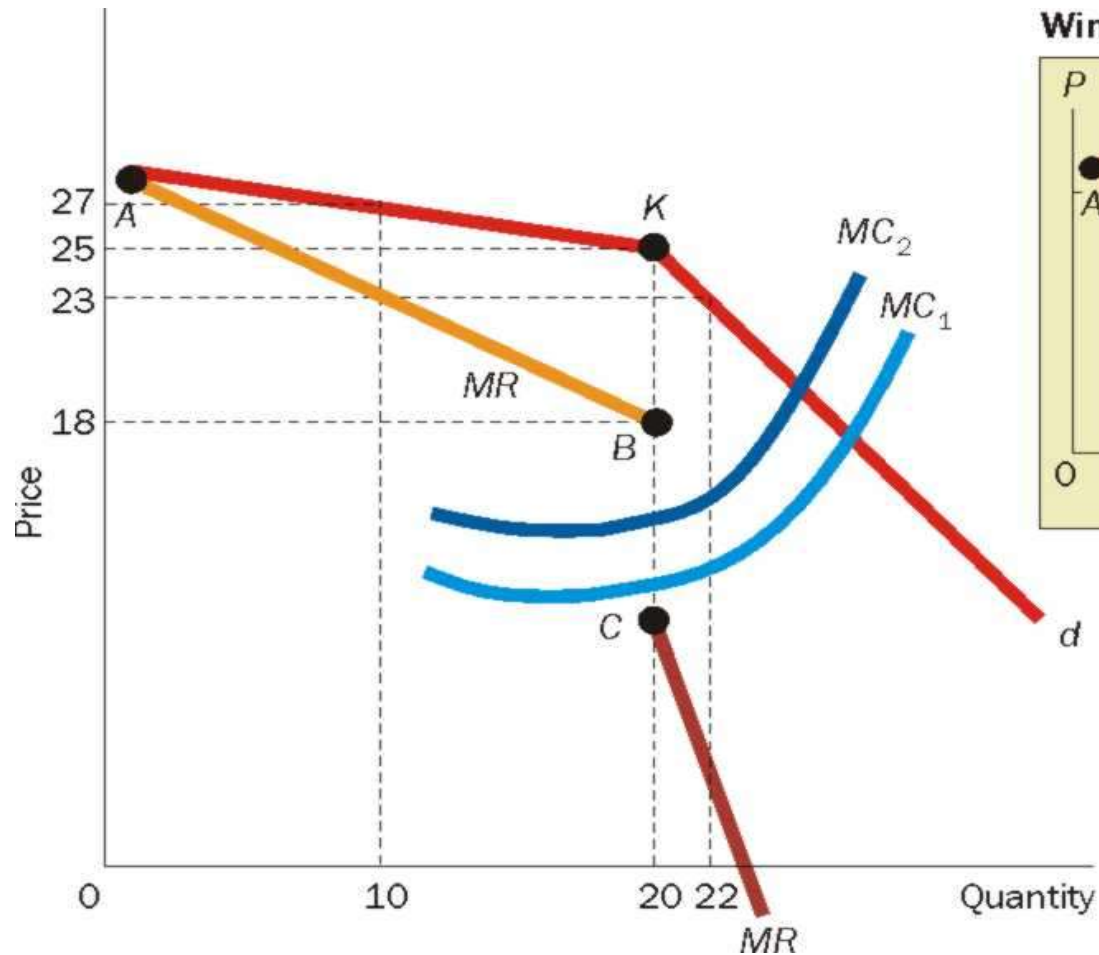
- The situation for a representative firm of the cartel: in long-run competitive equilibrium, it produces q_1 and charges P_1 , earning zero economic profits. As a consequence of the cartel agreement, it reduces output to q_c and charges P_c . Its profits are the are CP_cAB . If it cheats on the cartel agreement and others do not, the firm will increase output to q_{cc} and reap profits of FP_cDE .

As long as other firms do not cheat, this firm views its demand curve as horizontal at the cartel price (P_c) because it is one of a number of firms, so it cannot affect price by changing output. Therefore it can produce and sell additional units of output without lowering price.

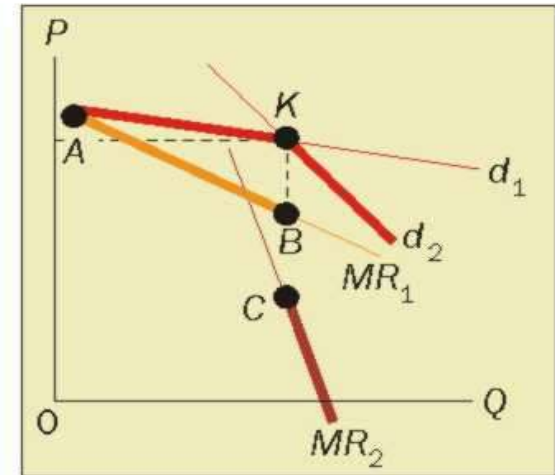
The Kinked Demand Curve Theory

- The key behavioral assumption is that if a single firm lowers price, other firms will do likewise, but if a single firm raises price, other firms will not follow suit.

The Kinked Demand Curve Theory



Window



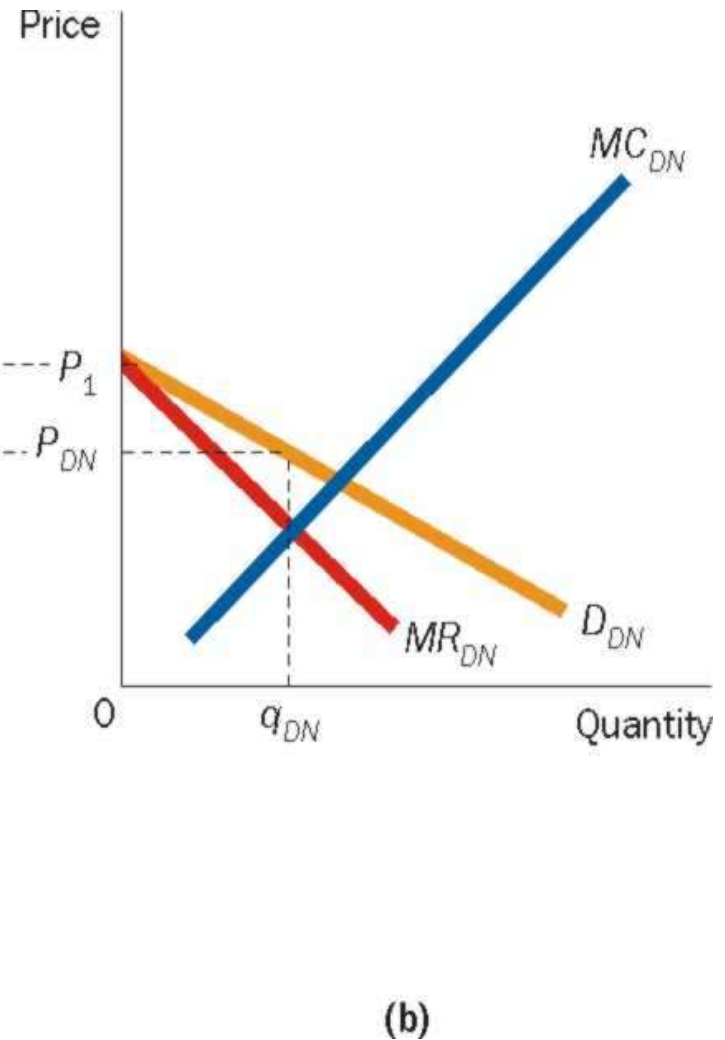
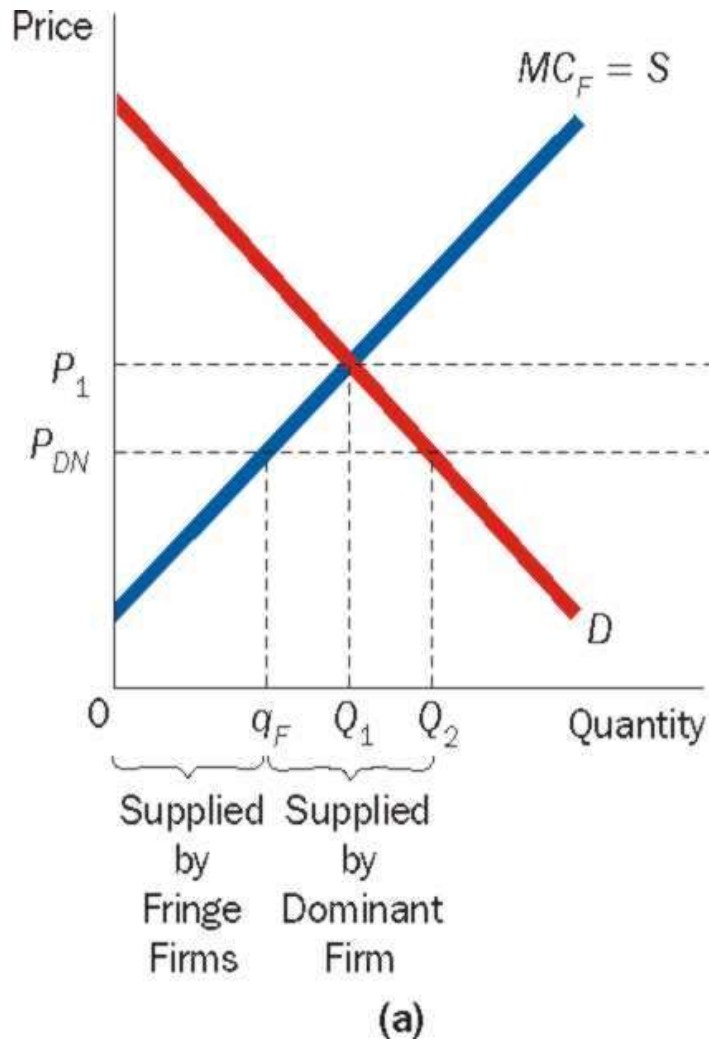
Observations About Kinked Demand Theory

- Prices are “sticky” if oligopolistic firms face kinked demand curves. Costs can change within certain limits, and such firms will not change their prices because they expect that none of their competitors will follow their price hikes, but that all will match their price cuts.
- The kinked demand curve posits that prices in oligopoly will be less flexible than in other market structures.

Price Leadership Theory

- One firm in the industry, called the dominant firm, determines price and all other firms take this price as given.
- The dominant firm sets the price that maximizes its profits, and all other firms take this price as given.
- All other firms are seen as price takers. They will equate price with their respective marginal costs.

Price Leadership Theory



- There is one dominant firm and a number of fringe firms. The horizontal sum of the marginal cost curves of the fringe firms is the supply curve. At P_1 , the fringe firms supply the entire market. The dominant firm derives its demand curve by computing the difference between market demand, D , and MC_F at each price below P_1 . It then produces q_{DN} and charges P_{DN} . P_{DN} becomes the price that the fringe firms take. They equate price and marginal cost and produce q_F in (a). The remainder of the output is produced by the dominant firm.

Theory of Contestable Markets

- There is easy entry into the market and costless exit from the market.
- New firms entering the market can produce the product at the same cost as existing firms.
- Firms exiting the market can easily dispose of their fixed assets by selling them elsewhere.

Criticism of Contestable Markets

Criticized because of the assumption there is free entry into and costless exit from the industry.

Conclusions of Contestable Markets

- Even if an industry is composed of a small number of firms, or simply one firm, this is not evidence that the firms perform in a noncompetitive way.
- Profits can be zero in an industry even if the number of sellers in the industry is small.
- If a market is contestable, inefficient producers cannot survive.
- A contestable market encourages firms to produce at their lowest possible average total cost and charge $P=ATC$, it follows that they will also sell at a price equal to marginal cost.